FLEET PERFORMANCE & DELIVERY EFFICIENCY DASHBOARD – SUMMARY REPORT

Project Overview

This dashboard was developed using Power BI to analyze and monitor the fleet performance of a logistics company. The dataset was cleaned, transformed, and modeled before creating interactive visualizations. The dashboard provides insights into on-time delivery performance, fuel efficiency, cost per km, and route-wise delivery analysis.

Data Preparation

- Imported the dataset (logistics_cleaned_dataset.xlsx) into Power BI with Trip_Data and Vehicle_Master sheets.
- Identified and removed the unwanted 'Projects' column containing portfolio content and null values.
- Added Index column and generated Trip_ID values (T01 to T50) using Transform > Prefix step.
- Replaced erroneous Distance_km value '116+666' with correct value 782 using Replace Values.
- Changed Distance_km column data type from Text to Decimal Number to enable DAX calculations.
- Filled missing Fuel_Consumed_L values using average fuel consumption per Vehicle Type.
- Created Many-to-One relationship between Trip_Data[Vehicle_ID] and Vehicle_Master[Vehicle_ID].

DAX Measures Created

- Fuel Efficiency = DIVIDE(SUM(Trip_Data[Distance_km]), SUM(Trip_Data[Fuel_Consumed_L])) — km per litre
- On Time Delivery % = DIVIDE(COUNTROWS(FILTER(Trip_Data, Delivery_Status = "On-Time")), COUNTROWS(Trip_Data))
- Cost per km = DIVIDE(SUM(Trip_Data[Distance_km]) * 90, SUM(Trip_Data[Fuel_Consumed_L])) — fuel cost per km

Key Dashboard Metrics

- Total Trips: 50 | On-Time Trips: 30 | Late Trips: 20
- On-Time Delivery Rate: 60%
- Overall Fuel Efficiency: 11.58 km/L
- Cost per km: Approx. Rs. 1,042
- Unique Vehicles: 7 | Unique Drivers: 10 | Unique Routes: 6 cities

Key Insights

- 60% of trips were delivered on time — Mumbai and Kolkata routes show the highest on-time delivery performance.
- Fuel efficiency declined slightly from January to February 2023 — vehicle maintenance check recommended.
- Chennai and Bangalore origin routes show higher late delivery counts — route optimization needed.
- Mini-Truck (V04) and Truck vehicles cover longer distances but have higher fuel consumption.
- Overall cost per km is high — fuel-efficient routing can significantly reduce operational costs.

Visualizations Used

- KPI Cards — On-Time Delivery Rate and Cost per km
- Bar Chart — On-Time Delivery % by Route (Origin city)
- Line Chart — Fuel Efficiency trend by Month (January to February 2023)
- Map Visual — Delivery performance by Origin city across India
- Funnel Chart — On-Time (30) vs Late (20) delivery count comparison
- Slicers — Vehicle Type, Delivery Status, and Origin City filters for interactive analysis

Conclusion

The Fleet Performance & Delivery Efficiency Dashboard enables effective monitoring of trip operations, fuel usage, and route performance. The interactive dashboard helps logistics managers identify underperforming routes, optimize vehicle allocation, and improve on-time delivery rates through data-driven decision-making. Key recommendations include route optimization for late-delivery cities, regular vehicle maintenance to sustain fuel efficiency, and driver-level performance tracking to achieve an 85%+ on-time delivery target.